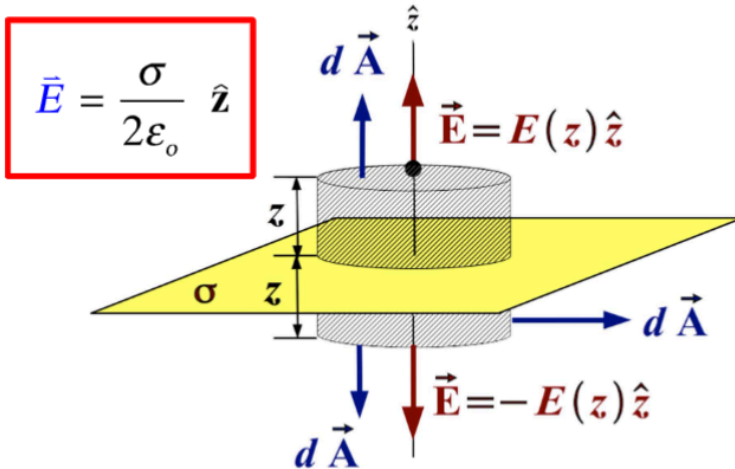
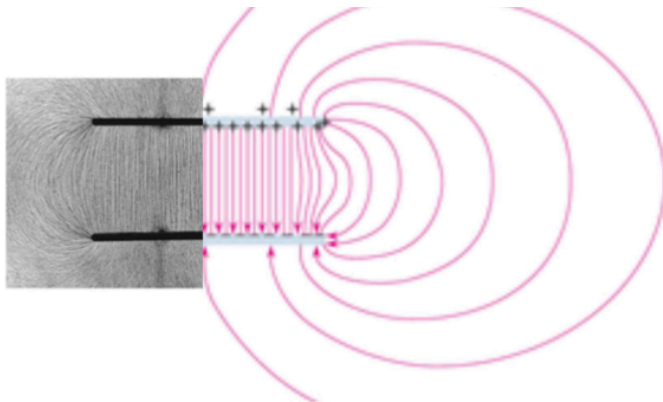
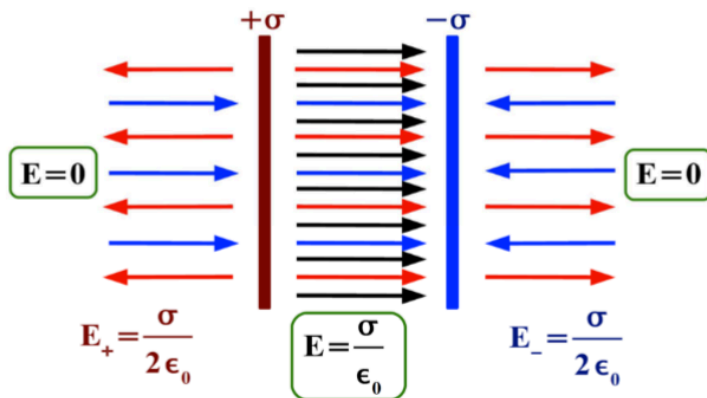


Lecture 4 Conductors, Capacitors, and Insulators



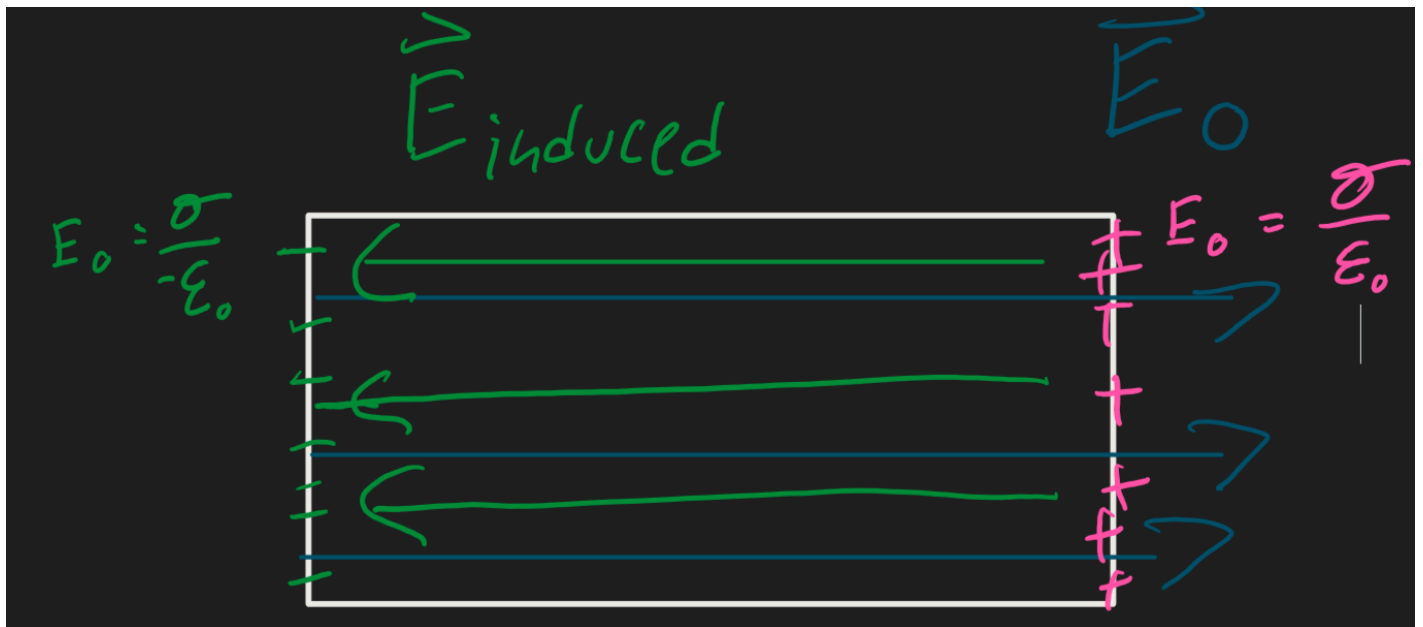
Example: Two Infinite Uniformly Charged Plates



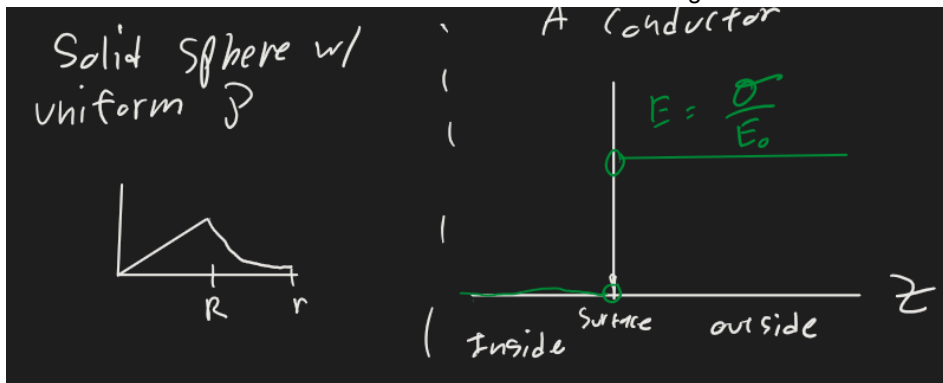
Finite plates → "fringing fields"

Conductors: A material where the charge is free to move around. In equilibrium, the field in a conductor is zero.

If there is an electric field on a conductor, the electrons will move to the edge of the conductor and keep moving until the new field by the uneven charge distribution cancels out the original field. They will do keep moving along the field lines until they make the net electric field on the surface of the conductor is zero.



The electric field is discontinuous when there is a surface charge.



How do conductors compare to dielectric materials, aka insulators.

Dielectrics are just bad conductors. There are still charges that feel the effect of a field, but they can't move well.

In polar dielectrics materials, the electrons are bound to their molecules and can't move as in a metal, so the molecules rotate and align with the magnetic field. The internal induced electric field is not sufficient to perfectly cancel the external electric field.

In non-polar dielectric materials, there is an induced dipole that then makes the material behave similarly to a polar dielectric.

The dielectric constant is

$$k_e = \frac{E_0}{E} = \frac{E_0}{E_0 - E_{\text{ind}}}$$

$$\vec{E}_{\text{tot}} = \frac{\vec{E}_0}{\epsilon}$$

A large ϵ corresponds to a conductor, a small one is an insulator.

In a vacuum, the electric field would not be reduced at all, so $\epsilon = 1$ with no material.

See previous: [Lecture 3 Gauss's law](#)

See next: [Differential Form of Gauss's Law](#)